

**Instructor:** Peter Mao

**Office:** Korman 209

**Email:** pm929@drexel.edu

**MRC Hours:** 1:00 - 3:00 on Tuesdays and 2:00 - 5:00 on Fridays

**Office Hours:** 12:00 - 1:00 on Tuesdays (Korman 209) or by appointment

**Main Text:** *Discrete Mathematics: An Open Introduction, 3rd edition* by Oscar Levin

**Supplementary Texts:** *Number Theory: In Context and Interactive* by Karl-Dieter Crisman

*Applied Combinatorics* by Mitchel T. Keller and William T. Trotter

Note that all three of these textbooks are meant to be accessed online for full functionality. The main textbook has solutions and/or hints to most of the exercises therein, so they can serve as a source of comparison or inspiration for homework problems. The other two textbooks both take advantage of live code evaluation which can help to make some things more concrete.

### **Course Description:**

Elementary set theory, combinatorics, elementary number theory, graphs, and special topics chosen from formal language theory, graph algorithms, coding theory, and other applications.

### **Pre-Requisites:**

MATH 220 Minimum Grade: C or CS 270 Minimum Grade: D or ECE 200 Minimum Grade: D

### **Homework**

Homework will be assigned on a weekly basis. It will be comprised of two parts: WeBWorK questions and written questions. The WeBWorK questions will be drawn from the textbook and submitted automatically via WeBWorK. The written questions will be primarily proof-based and they should be submitted via Gradescope. Preferably solutions can be written up in LaTeX or written digitally, but taking a legible photo of your work on paper is also acceptable. You may consult with your fellow classmates, me, and other resources (like the Math Resource Center in Korman 207), but your written submissions should be in your own words and your sources and those who you consulted with should be cited.

### **Homework Corrections**

For written homework, students may submit corrections to their work for the week following the initial grading (usually one week from the deadline) except for week 4 and week 9 (so that solutions can be posted before the midterm exam and final exam respectively). There is no grade penalty for submitting corrections and the better answer between the original and corrected version will be taken.

### **Exams**

Exams will be in a timed take-home format. Specifically, a timer will start upon you accessing the test and you will be required to submit before the timer concludes. It is take-home in the sense that, within the window of time (a week), you can choose when is most convenient for you to take the exam. Unlike with homework, consulting others is not allowed for exams, but referring to the textbooks, your notes, or homework solutions is allowed. There will be two such exams, a midterm and a final. These will be assigned in Week 6 and Week 10 with the normal week from the assignment to start them.

**Grade Breakdown**

- WeBWork Homework: 10%
- Written Homework: 40%
- Midterm Exam: 25%
- Final Exam: 25%

**Grade Scale**

|    |        |   |       |    |       |
|----|--------|---|-------|----|-------|
| A+ | 96-100 | A | 91-95 | A- | 86-90 |
| B+ | 82-85  | B | 78-81 | B- | 74-77 |
| C+ | 70-73  | C | 66-69 | C- | 62-65 |
| D+ | 56-61  | D | 51-55 | F  | <50   |

**Attendance**

Students are expected to attend lecture and are responsible for all material covered in class.

**Lateness**

Late written assignments may be subject to a 5% grade penalty for each lecture late.

**Academic Policies**

This course adheres strictly to the university's policies on the following:

- Academic Dishonesty
- Course Add/Drop Policy
- Course Withdrawal Policy
- Incomplete Grade Policy
- Grade Appeals
- Code of Conduct

**Disability Accommodations**

It is the University's policy to provide a learning environment that provides all students with the opportunity to realize their full potential. To this end, the goal of the Office of Disability Resources ("ODR") is to assist students with disabilities in becoming self-sufficient, independent, and successful members of the University's community and to provide students with disabilities with equal opportunity of access to university courses, programs, facilities, services, and activities.

Students requesting accommodations due to a disability at Drexel University need to request a current Accommodations Verification Letter (AVL) in the ClockWork Database before accommodations can be made. These requests are received by Disability Resources (DR), who then issues the AVL to the appropriate contacts. For additional information, visit the website, or contact the ODR office by email [disability@drexel.edu](mailto:disability@drexel.edu), or by phone 215-895-1401.

**Changes to Syllabus**

Any changes to this syllabus will be communicated in class and by email.